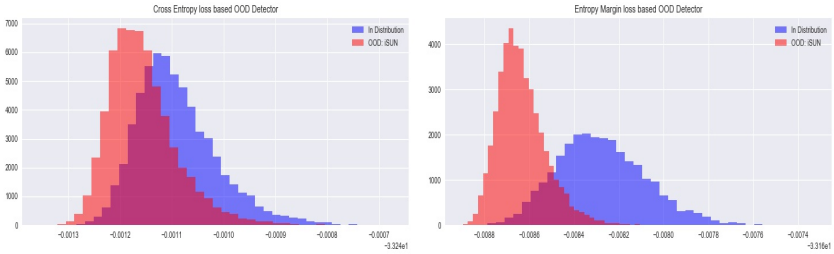
OOD Dataset	DenseNet-BC CIFAR-10	DenseNet-BC CIFAR-100	WRN-28-10 CIFAR-10	WRN-28-10 CIFAR-100
TINc	1.49 ± 0.23	10.26 ± 1.33	1.3 ± 0.33	10.35 ± 2.21
TINr	3.95 ± 0.75	26.58 ± 4.16	4.56 ± 1.29	29.84 ± 5.12
LSUNc	4.54 ± 1.42	16.95 ± 1.27	3.81 ± 1.22	15.51 ± 1.4
LSUNr	1.3 ± 0.6	20.22 ± 2.79	1.53 ± 0.41	22.51 ± 6.08
UNFM	14.37 ± 31.84	38.79 ± 19.41	0.75 ± 1.24	47.67 ± 47.19
GSSN	27.09 ± 40.02	82.24 ± 12.81	31.47 ± 33.95	67.48 ± 44.33

Table 4: Mean and standard deviation of FPR at 95% TPR



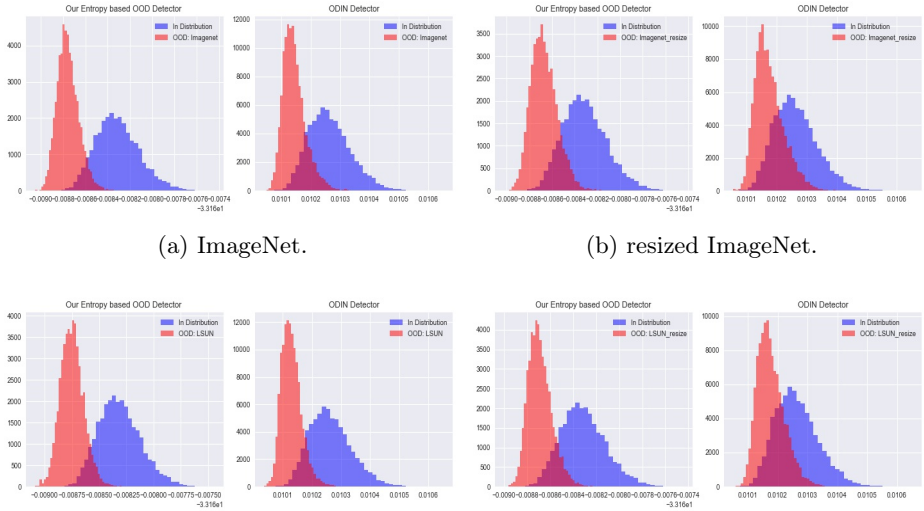
(a) Cross entropy loss. (b) Margin entropy loss.

Fig. 1: Histogram of ID and OOD detection scores of the proposed method and ODIN [13]

with only cross entropy loss, while Figure 1(b) is trained with proposed margin entropy loss and cross entropy, the proposed OOD detector is used in both bases. As shown in Figure 1, the proposed margin entropy loss helps to better separate ID and OOD distributions than using cross entropy loss alone. Figure 2 presents the histogram plot of OOD detection scores on ID and OOD samples for both our method and ODIN [13]. As shown in Figure 2, the proposed method has less overlap between OOD samples and ID samples compared to ODIN [13] and thus separates ID and OOD distributions better.

5 Conclusion and Future Work

As deep learning is widely adopted in many commercially important applications, it is very important that anomaly detection algorithms are developed for these algorithms. In this work, we have proposed an anomaly detection algorithm for deep neural networks which is an ensemble of leave-out-classifiers. These classifiers are learned by maximizing the margin-loss between the entropy of OOD samples and in-distribution samples. A random subset of training data serves as OOD data while the rest of the data serves as in-distribution. We show our algorithm significantly outperforms the current state-of-art methods [7], [12] and [13] across almost all the benchmarks. Our method contains three important components, novel loss function, the ensemble of leave-out classifiers, and novel



(a) ImageNet. (b) resized ImageNet.

(c) LSUN. (d) resized LSUN.

Fig. 2: Histogram of ID and OOD detection scores with proposed OOD detector v.s. ODIN OOD detector

out-of-distribution detector. Each of this component improves OOD detection performance. Each of them can be applied independently on top of other methods.

We also note that this method opens up several directions of research to pursue. First, the proposed method of the ensemble of neural networks requires large memory and computational resources. This can potentially be alleviated by all the networks sharing most of the parameters and branch away individually. Also, the number of splits can be used to trade off between detection performance and computational overhead. Notice that based on ablation study (Table 2) and detailed 3 splits results in supplementary document, even 3 splits outperform ODIN [13]. For use cases where reducing computational time is critical, we recommend to use 3 splits. Please see supplementary material for detailed results on 3 splits. Our current work requires an OOD dataset for hyper-parameter search. This problem can potentially be solved by investigating other surrogate functions for entropy which are better behaved with the epochs.

References

1. Bendale, A., Boult, T.E.: Towards open world recognition. In: IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 1893–1902 (2015)
2. Bendale, A., Boult, T.E.: Towards open set deep networks. In: IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 1563–1572 (2016)
3. Chandola, V., Banerjee, A., Kumar, V.: Anomaly detection: A survey. ACM Comput. Surv. **41**(3), 15:1–15:58 (2009)

4. Fujimaki, R., Yairi, T., Machida, K.: An approach to spacecraft anomaly detection problem using kernel feature space. In: KDD. pp. 401–410 (2005)
5. Goadrich, M., Oliphant, L., Shavlik, J.: Creating ensembles of first-order clauses to improve recall-precision curves. Machine Learning **64**, 231–262 (2006)
6. Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., Bengio, Y.: Generative adversarial nets. In: Ghahramani, Z., Welling, M., Cortes, C., Lawrence, N.D., Weinberger, K.Q. (eds.) Advances in Neural Information Processing Systems (NIPS) (2014)
7. Hendrycks, D., Gimpel, K.: A baseline for detecting misclassified and out-of-distribution examples in neural networks. In: International Conference on Learning Representations (ICLR) (2017)
8. Huang, G., Liu, Z., Weinberger, K.Q.: Densely connected convolutional networks (2016), arXiv preprint arXiv:1608.06993
9. https://tiny imagenet.herokuapp.com:
10. Krizhevsky, A., Hinton, G.: Learning multiple layers of features from tiny images
11. LeCun, Y., Chopra, S., Hadsell, R., Ranzato, M., Huang, F.: A tutorial on energy-based learning. In: Predicting structured data (2006)
12. Lee, K., Lee, H., Lee, K., Shin, J.: Training confidence-calibrated classifiers for detecting out-of-distribution samples. In: International Conference on Learning Representations (ICLR) (2018), `https://openreview.net/forum?id=ryiAv2xAZ`
13. Liang, S., Li, Y., Srikant, R.: Enhancing the reliability of out-of-distribution image detection in neural networks. In: International Conference on Learning Representations (ICLR) (2018)
14. Liu, F.T., Ting, K.M., Zhou, Z.: Isolation forest. In: ICDM. pp. 413–422 (2008)
15. Lu, W., Traoré, I.: Unsupervised anomaly detection using an evolutionary extension of k-means algorithm. IJICS **2**(2), 107–139 (2008)
16. Phua, C., Alahakoon, D., Lee, V.C.S.: Minority report in fraud detection: classification of skewed data. SIGKDD Explorations **6**(1), 50–59 (2004)
17. Rudd, E.M., Jain, L.P., Scheirer, W.J., Boult, T.E.: The extreme value machine. IEEE Trans. Pattern Anal. Mach. Intell. **40**(3), 762–768 (2018)
18. Scheirer, W.J., Jain, L.P., Boult, T.E.: Probability models for open set recognition. IEEE Trans. Pattern Anal. Mach. Intell. **36**(11), 2317–2324 (2014)
19. Scheirer, W.J., de Rezende Rocha, A., Sapkota, A., Boult, T.E.: Toward open set recognition. IEEE Trans. Pattern Anal. Mach. Intell. **35**(7), 1757–1772 (2013)
20. Xu, P., Ehinger, K.A., Zhang, Y., Finkelstein, A., Kulkarni, S.R., Xiao, J.: Turkergaze: Crowdsourcing saliency with webcam based eye tracking (2015), arXiv preprint arXiv:1504.06755
21. Yu, F., Zhang, Y., Song, S., Seff, A., Xiao, J.: Lsun: Construction of a large- scale image dataset using deep learning with humans in the loop (2015), arXiv preprint arXiv:1506.03365
22. Zagoruyko, S., Komodakis, N.: Wide residual networks (2016), arXiv preprint arXiv:1605.07146